UPAR Challenge 2027 · Track 1 · WACV 2027 workshop

Pedestrian Attribute Recognition on Unseen Cameras

Ibrahim Johar Farooqi (23K-0074) · Syed Ukkashah Ahmed (23K-0055)

Computer Vision, Fall 2026 · FAST-NUCES Karachi

The task

40

yes/no attributes per cropped person image

Gender, age, hair, clothing type and colour, bags, hats, glasses

Training data

Market1501, PA-100K, PETA: 97,669 train / 33,407 validation images

Hidden test set

29,248 images from cameras the model has never seen

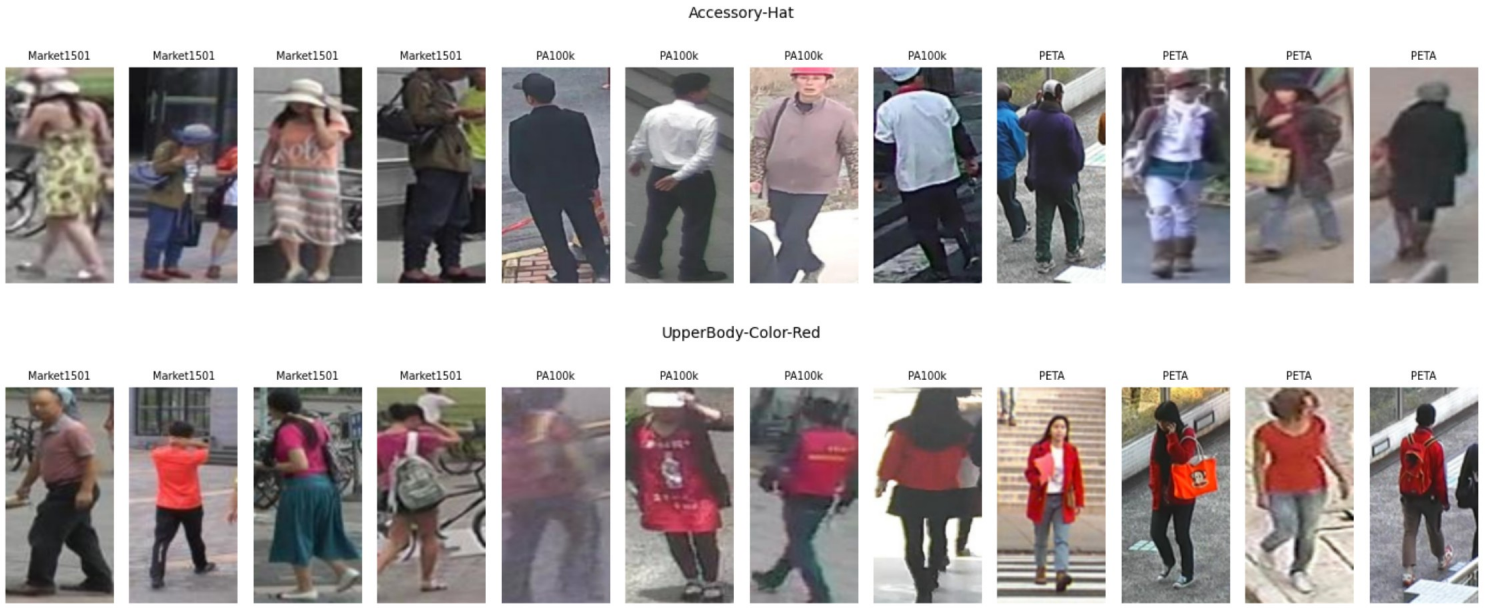
Submission

Code, run offline on the organisers' A10 GPU within 5 minutes

Score

Harmonic mean of mA and instance F1

What the data looks like



The same label looks different in each dataset: sun hats vs caps, shades of red, image quality.

21 / 40

attributes are positive in under 5% of images

24 / 40

attributes are colours, so no hue augmentation

Approach



One training recipe for every backbone

Weighted BCE · dataset-balanced sampling · AdamW · 6 epochs · 2× T4 on Kaggle

Comparing backbones: CNN vs transformer

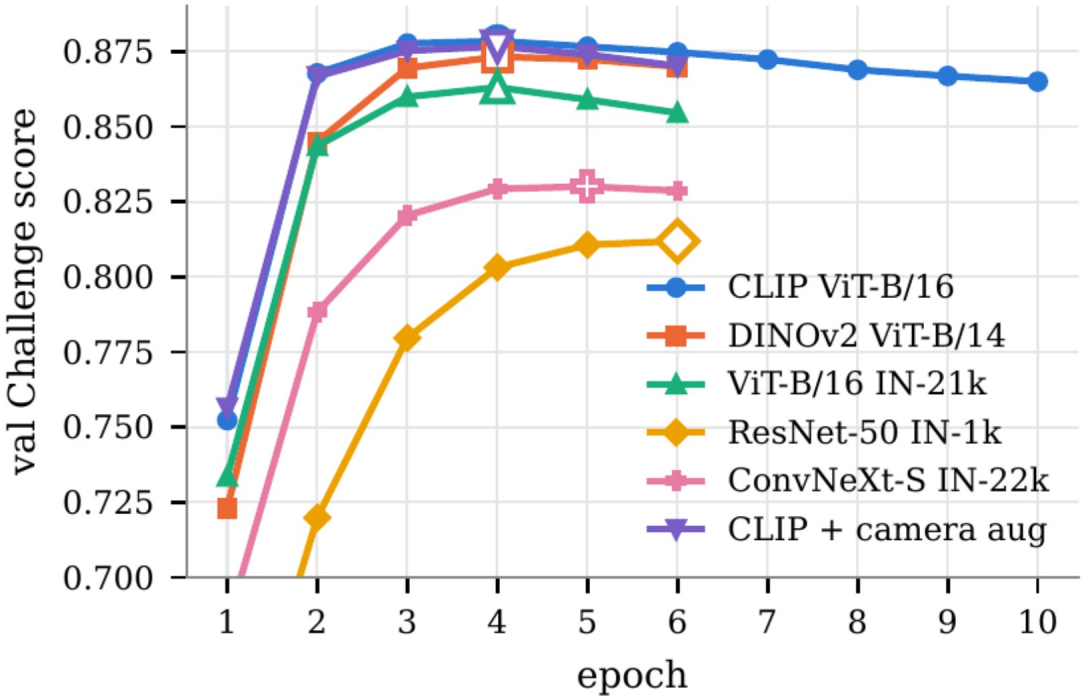
Backbone	Type	Val score
ResNet-50 (ImageNet-1k)	CNN	0.826
ConvNeXt-S (ImageNet-22k)	CNN	0.845
ViT-B/16 (ImageNet-21k)	Transformer	0.873
DINOv2 ViT-B/14	Transformer	0.890
CLIP ViT-B/16	Transformer	0.891

+0.028

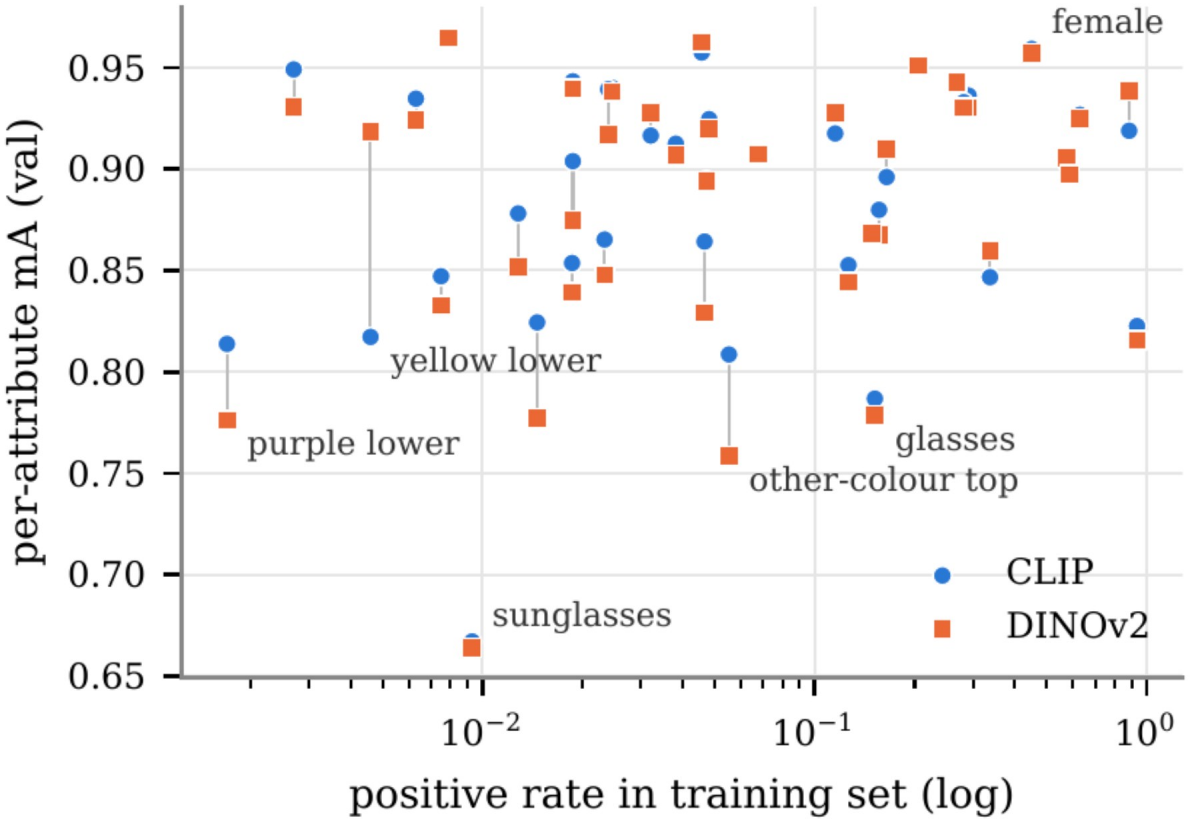
Transformer over CNN, same pretraining (better on 39 of 40 attributes)

+0.018

from broader pretraining on top



Two models, different mistakes



DINOv2 is better on

Shape and carried objects: bags, backpacks, skirts

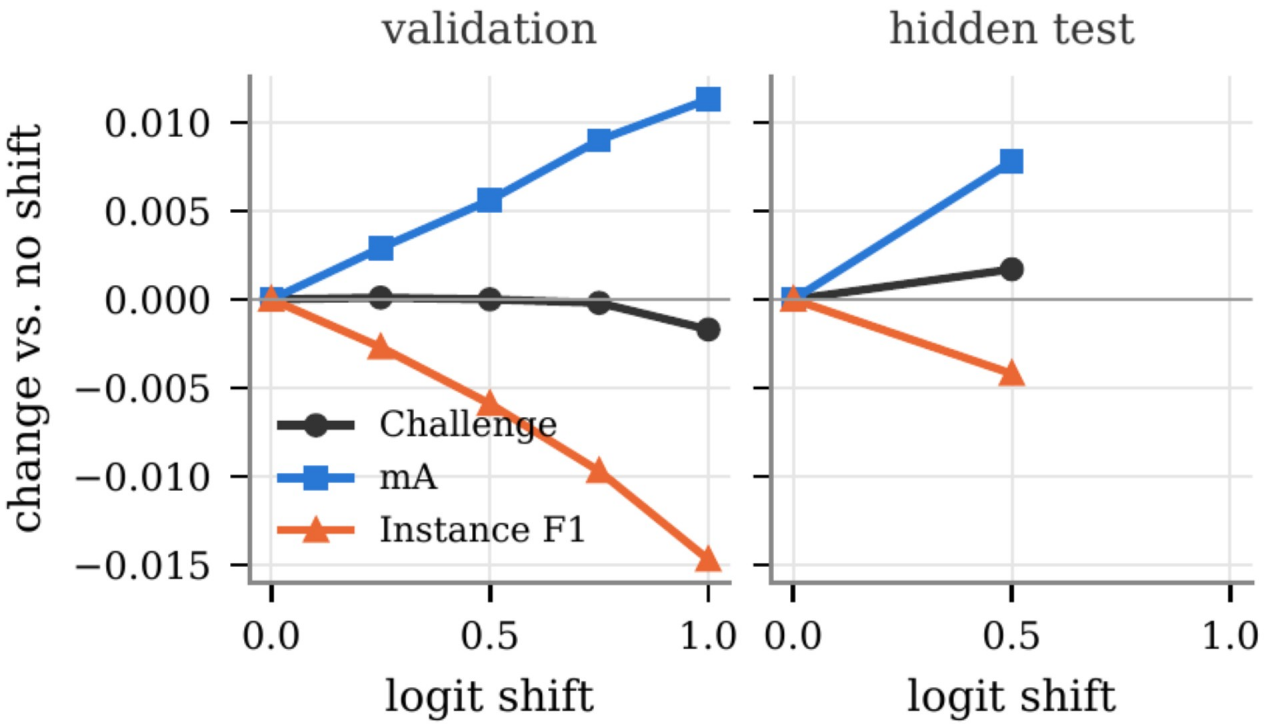
CLIP is better on

Colours, age and baldness

0.7766 → 0.7816

Averaging the two: the biggest gain on the test set

Validation vs the unseen test cameras



0.89 → 0.78

Validation vs test: a large domain gap

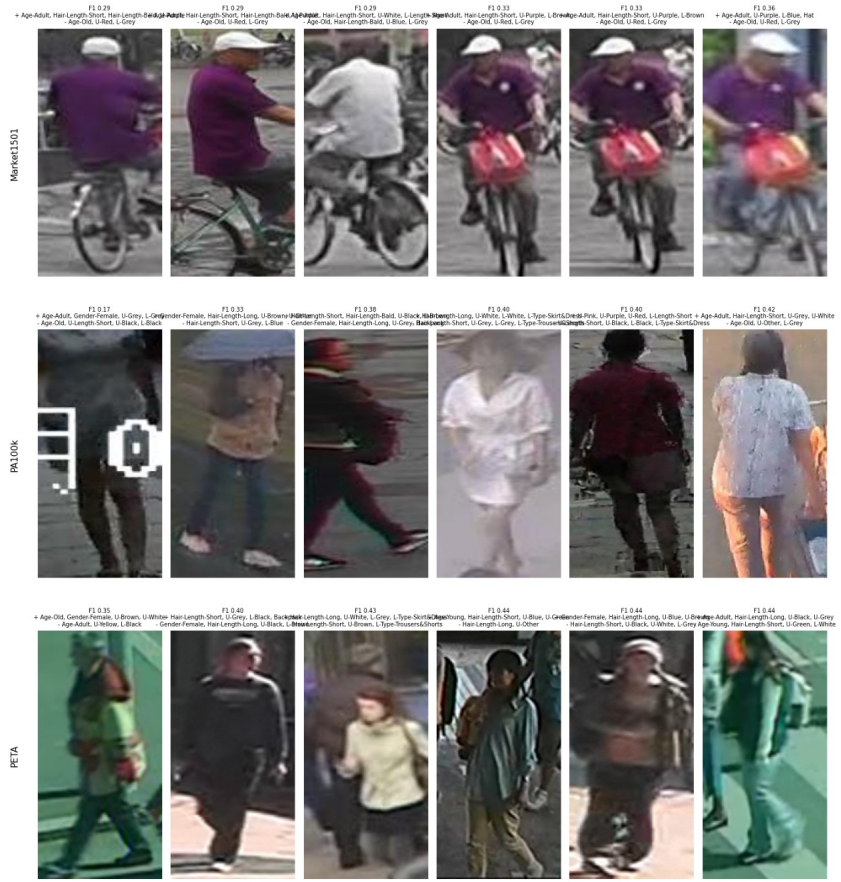
On new cameras, "yes" too rarely

Test recall 0.82 vs 0.92 on validation

Lower every threshold slightly

+0.5 logit shift: no change on validation, test rises to 0.7833

Where it still fails



0.67

mA on sunglasses

Small objects

Sunglasses are a few pixels in a low-resolution crop

Rare colours

Few positive examples to learn from

Noisy labels

The worst Market1501 images are one cyclist whose purple shirt is labelled red

Demo and final result



The submitted model on validation images

Also tried, did not help

Retraining on train + val (−0.009) · simulated camera blur/JPEG (−0.007)

0.7833

mA 0.7873 · instance F1 0.7794

Rank 35 on 20 October 2026

26	murtazajafri	2026-10-20 19:21	1005584	1005584_VID-Docker-GPU	**0.7884**	0.7937	0.575	0.671	0.7491	0.8349	0.7833
27	syouya_tobita	2026-10-20 16:16	1005192	1005192_VID-Docker-GPU	**0.7873**	0.802	0.5678	0.6567	0.7317	0.8369	0.7731
28	aio_pentatonis	2026-10-19 22:04	1003491	1003491_VID-Docker-GPU	**0.787**	0.789	0.5668	0.6737	0.7525	0.8385	0.785
29	yudongtian	2026-10-13 06:08	981477	981477_VID-Docker-GPU	**0.7866**	0.7886	0.5834	0.6733	0.7634	0.8193	0.7847
30	gadgadgad	2026-10-20 07:46	1004240	1004240_VID-Docker-GPU	**0.7861**	0.7916	0.5861	0.666	0.7423	0.8384	0.7807
31	huethuong	2026-10-17 06:38	994799	994799_VID-Docker-GPU	**0.786**	0.7923	0.5853	0.6644	0.7383	0.841	0.7797
32	VisionEncoders	2026-10-12 15:09	980151	980151_VID-Docker-GPU	**0.7856**	0.7864	0.6049	0.6721	0.7508	0.8364	0.7848
33	tienptx	2026-10-09 08:18	970807	970807_VID-Docker-GPU	**0.7853**	0.7943	0.5539	0.661	0.7376	0.8359	0.7765
34	Nguyen Duc Thanh	2026-10-20 10:07	1004456	1004456_VID-Docker-GPU	**0.7847**	0.7803	0.6019	0.6778	0.7555	0.8398	0.789
35	ibrahimjohar	2026-10-20 20:41	1005773	1005773_VID-Docker-GPU	**0.7833**	0.7873	0.5907	0.6644	0.7401	0.8385	0.7794
36	pa_pa_pa_pa	2026-10-20 10:30	1004489	1004489_VID-Docker-GPU	**0.7827**	0.7853	0.5737	0.6667	0.7456	0.8322	0.78
37	be_tap_code	2026-10-20 12:16	1004689	1004689_VID-Docker-GPU	**0.7827**	0.7853	0.5737	0.6667	0.7456	0.8322	0.78

Lessons and next steps

What we learned

Broad pretraining and transformers matter most; ensembles of differently pretrained models help

Validation ranks ideas but does not predict test scores on new cameras

Next

Higher resolution for small accessories

Validate on a held-out dataset

Distil the ensemble into one model

Code: github.com/ibrahimjohar/upar27-challenge-pedestrian-attributes